6. [Maximum mark: 6]

The line L_1 has vector equation $\boldsymbol{r} = 4\boldsymbol{i} - \boldsymbol{k} + \lambda(a\boldsymbol{j} + \boldsymbol{k})$, where $a, \lambda \in \mathbb{R}$.

The line L_2 has vector equation $\boldsymbol{r} = \boldsymbol{i} - b\boldsymbol{k} + \mu(\boldsymbol{i} + 2\boldsymbol{j} + 3\boldsymbol{k})$, where $b, \mu \in \mathbb{R}$.

The lines L_1 and L_2 are perpendicular and intersect at a unique point.

Find the value of a and the value of b.

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7. [Maximum mark: 5]

Consider the complex number $z = 3^{i-1}$.

(a) Write the integer 3 in the form e^a where $a \in \mathbb{R}$. [1]

(b) Hence, giving your answers in the form $p\cos(\ln q)$ where $p, q \in \mathbb{Q}^+$, find

(i) $\mathrm{Re}(z)$;

(ii) $\mathrm{Re}\left(\frac{1}{z}\right)$. [4]

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2. [Maximum mark: 5]

The line L_1 is defined by the Cartesian equation $\frac{x-1}{2} = \frac{y+2}{3} = z$.

(a) Find a vector equation of L_1. [2]

A second line L_2 is defined by the vector equation $\boldsymbol{r} = \begin{pmatrix} 0 \\ 4 \\ -8 \end{pmatrix} + t \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$, where $t \in \mathbb{R}$.

(b) Find the coordinates of the point where L_1 and L_2 intersect. [3]

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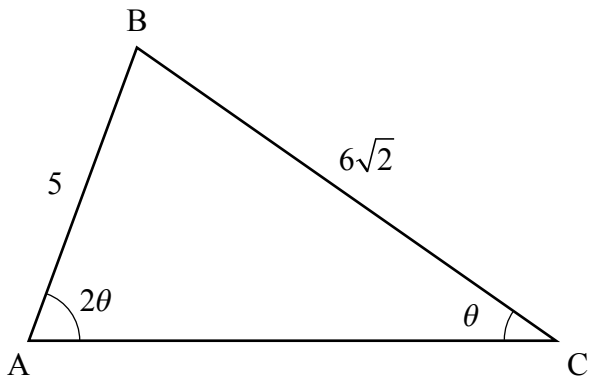
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3. [Maximum mark: 7]

The following diagram shows a non-right angled triangle ABC.

diagram not to scale



$\mathrm{AB} = 5$, $\mathrm{BC} = 6\sqrt{2}$, $\mathrm{A\hat{C}B} = \theta$ and $\mathrm{B\hat{A}C} = 2\theta$, where $0 < \theta < \frac{\pi}{2}$.

(a) Using the sine rule, show that $\cos\theta = \frac{3\sqrt{2}}{5}$. [3]

(b) Hence, find $\sin\theta$. [2]

Point D is located on $[\mathrm{AC}]$ such that the area of triangle BCD is $2\sqrt{14}$.

(c) Find DC. [2]

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10. [Maximum mark: 19]

The point $\mathrm{P}(-1, 1, -13)$ lies on the line L_1. The line L_1 has a direction vector $\begin{pmatrix} 7 \\ 1 \\ 2 \end{pmatrix}$.

(a) Write down a vector equation for L_1 in the form $\boldsymbol{r} = \boldsymbol{a} + \lambda \boldsymbol{b}$. [1]

(b) Find a vector equation for line L_2 in the form $\boldsymbol{s} = \boldsymbol{c} + \mu \boldsymbol{d}$, given that L_2 passes through the points $\mathrm{A}(2, -4, 2)$ and $\mathrm{B}(7, -6, 1)$. [2]

(c) Show that L_1 and L_2 are skew. [5]

The point N lies on L_2.

(d) Find $\overrightarrow{\mathrm{PN}} \cdot \overrightarrow{\mathrm{AB}}$ in terms of μ. [4]

(e) Given that N is the point on L_2 that lies closest to P, find the coordinates of N. [3]

(f) Point O denotes the origin $(0, 0, 0)$.
Find the equation of the plane containing points O, P and N, giving your answer in the form $\alpha x + \beta y + \gamma z = \delta$, where $\alpha, \beta, \gamma, \delta \in \mathbb{Z}$. [4]

Do **not** write solutions on this page.

11. [Maximum mark: 20]

(a) It is given that $w = 1 + \mathrm{i}\sqrt{3}$.

(i) Express w in the form $w = r\mathrm{e}^{\mathrm{i}\alpha}$, where $r > 0$ and $-\pi < \alpha \leq \pi$.

(ii) Show that $\left(1 + \mathrm{i}\sqrt{3}\right)^8 + \left(1 - \mathrm{i}\sqrt{3}\right)^8 = -256$. [6]

It is given that $\left(1 + \mathrm{i}\tan\theta\right)^n \equiv \dfrac{\cos n\theta + \mathrm{i}\sin n\theta}{\cos^n \theta}$ for $\cos\theta \neq 0$, $n \in \mathbb{Z}^+$.

(b) Show that $\left(1 + \mathrm{i}\tan\theta\right)^4 + \left(1 - \mathrm{i}\tan\theta\right)^4 \equiv \dfrac{2\cos 4\theta}{\cos^4 \theta}$. [3]

(c) Hence, determine the four roots of the equation $(1 + z)^4 + (1 - z)^4 = 0$, giving your answers in the form $z = \mathrm{i}\tan\left(\dfrac{k\pi}{8}\right)$, where $k \in \mathbb{Z}^+$. [4]

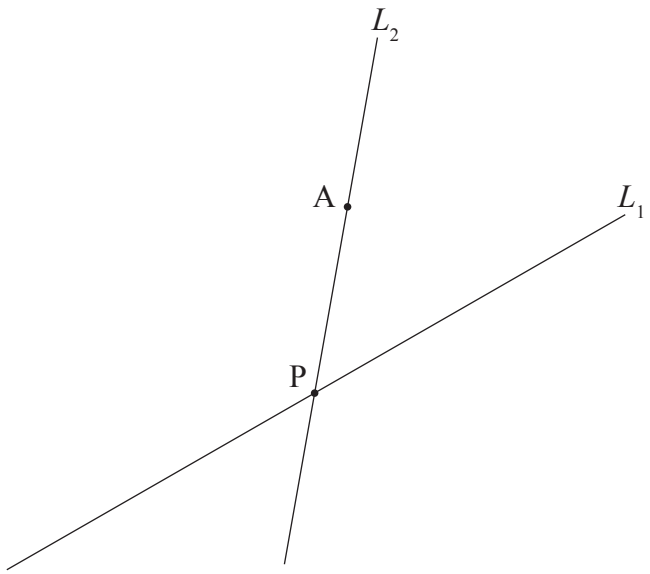
(d) By considering the binomial expansion of $(1 + z)^4 + (1 - z)^4$, show that $\tan\dfrac{\pi}{8} = \sqrt{3 - 2\sqrt{2}}$. [7]

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12. [Maximum mark: 21]

Two lines, L_1 and L_2, intersect at point P. Point $\mathrm{A}(2t, 8, 3)$, where $t > 0$, lies on L_2. This is shown in the following diagram.

diagram not to scale



The acute angle between the two lines is $\frac{\pi}{3}$.

The direction vector of L_1 is $\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$, and $\overrightarrow{\mathrm{PA}} = \begin{pmatrix} 2t \\ 0 \\ 3+t \end{pmatrix}$.

(a) Show that $4t = \sqrt{10t^2 + 12t + 18}$. [4]

(b) Find the value of t. [4]

(c) Hence or otherwise, find the shortest distance from A to L_1. [4]

A plane, Π, contains L_1 and L_2.

(d) Find a normal vector to Π. [2]

The base of a right cone lies in Π, centred at A such that L_1 is a tangent to its base. The volume of the cone is $90\pi\sqrt{3}$ cubic units.

(e) Find the two possible positions of the vertex of the cone. [7]

References:

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Section B

Answer **all** questions in the answer booklet provided. Please start each question on a new page.

10. [Maximum mark: 18]

Consider the series $\ln x + p\ln x + \frac{1}{3}\ln x + \ldots$, where $x \in \mathbb{R}, x > 1$ and $p \in \mathbb{R}, p \neq 0$.

(a) Consider the case where the series is geometric.

(i) Show that $p = \pm\frac{1}{\sqrt{3}}$.

(ii) Hence or otherwise, show that the series is convergent.

(iii) Given that $p > 0$ and $S_\infty = 3 + \sqrt{3}$, find the value of x. [6]

(b) Now consider the case where the series is arithmetic with common difference d.

(i) Show that $p = \frac{2}{3}$.

(ii) Write down d in the form $k\ln x$, where $k \in \mathbb{Q}$.

(iii) The sum of the first n terms of the series is $\ln\left(\frac{1}{x^3}\right)$.

Find the value of n. [12]

11. [Maximum mark: 15]

Consider the three planes

$$\Pi_1 : 2x - y + z = 4$$
$$\Pi_2 : x - 2y + 3z = 5$$
$$\Pi_3 : -9x + 3y - 2z = 32$$

(a) Show that the three planes do not intersect. [4]

(b) (i) Verify that the point $\mathrm{P}(1, -2, 0)$ lies on both Π_1 and Π_2.

(ii) Find a vector equation of L, the line of intersection of Π_1 and Π_2. [5]

(c) Find the distance between L and Π_3. [6]

Do **not** write solutions on this page.

12. [Maximum mark: 21]

The function f is defined by $f(x) = e^x \sin x$, where $x \in \mathbb{R}$.

(a) Find the Maclaurin series for $f(x)$ up to and including the x^3 term. [4]

(b) Hence, find an approximate value for $\int_0^1 e^{x^2} \sin(x^2)\,dx$. [4]

The function g is defined by $g(x) = e^x \cos x$, where $x \in \mathbb{R}$.

(c) (i) Show that $g(x)$ satisfies the equation $g''(x) = 2\big(g'(x) - g(x)\big)$.

(ii) Hence, deduce that $g^{(4)}(x) = 2\big(g'''(x) - g''(x)\big)$. [5]

(d) Using the result from part (c), find the Maclaurin series for $g(x)$ up to and including the x^4 term. [5]

(e) Hence, or otherwise, determine the value of $\lim_{x \to 0} \dfrac{e^x \cos x - 1 - x}{x^3}$. [3]

References:

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11. [Maximum mark: 19]

Consider the line L_1 defined by the Cartesian equation $\frac{x+1}{2} = y = 3 - z$.

(a) (i) Show that the point $(-1, 0, 3)$ lies on L_1.

(ii) Find a vector equation of L_1. [4]

Consider a second line L_2 defined by the vector equation $\boldsymbol{r} = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} + t \begin{pmatrix} a \\ 1 \\ -1 \end{pmatrix}$,

where $t \in \mathbb{R}$ and $a \in \mathbb{R}$.

(b) Find the possible values of a when the acute angle between L_1 and L_2 is $45°$. [8]

It is given that the lines L_1 and L_2 have a unique point of intersection, A, when $a \neq k$.

(c) Find the value of k, and find the coordinates of the point A in terms of a. [7]

12. [Maximum mark: 20]

Let $f(x) = \sqrt{1+x}$ for $x > -1$.

(a) Show that $f''(x) = -\frac{1}{4\sqrt{(1+x)^3}}$. [3]

(b) Use mathematical induction to prove that $f^{(n)}(x) = \left(-\frac{1}{4}\right)^{n-1} \frac{(2n-3)!}{(n-2)!}(1+x)^{\frac{1}{2}-n}$

for $n \in \mathbb{Z}$, $n \geq 2$. [9]

Let $g(x) = \mathrm{e}^{mx}$, $m \in \mathbb{Q}$.

Consider the function h defined by $h(x) = f(x) \times g(x)$ for $x > -1$.

It is given that the x^2 term in the Maclaurin series for $h(x)$ has a coefficient of $\frac{7}{4}$.

(c) Find the possible values of m. [8]

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5. [Maximum mark: 4]

Given any two non-zero vectors, $\boldsymbol{a}$ and $\boldsymbol{b}$, show that $|\boldsymbol{a} \times \boldsymbol{b}|^2 = |\boldsymbol{a}|^2 |\boldsymbol{b}|^2 - (\boldsymbol{a} \cdot \boldsymbol{b})^2$.

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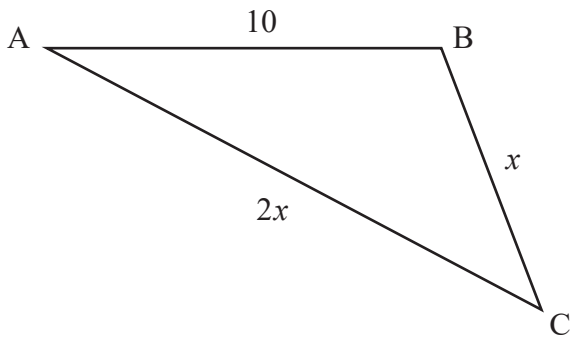
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6. [Maximum mark: 7]

The following diagram shows triangle ABC, with $\mathrm{AB} = 10$, $\mathrm{BC} = x$ and $\mathrm{AC} = 2x$.

diagram not to scale



Given that $\cos \hat{\mathrm{C}} = \frac{3}{4}$, find the area of the triangle.

Give your answer in the form $\frac{p\sqrt{q}}{2}$ where $p, q \in \mathbb{Z}^+$.

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8. [Maximum mark: 8]

The lines l_1 and l_2 have the following vector equations where $\lambda, \mu \in \mathbb{R}$.

$$l_1 : \boldsymbol{r}_1 = \begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -2 \\ 2 \end{pmatrix}$$

$$l_2 : \boldsymbol{r}_2 = \begin{pmatrix} 2 \\ 0 \\ 4 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$$

(a) Show that l_1 and l_2 do not intersect. [3]

(b) Find the minimum distance between l_1 and l_2. [5]

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9. [Maximum mark: 7]

By using the substitution $u = \sin x$, find $\int \frac{\sin x \cos x}{\sin^2 x - \sin x - 2} \mathrm{d}x$.

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12. [Maximum mark: 21]

The equations of two lines, L_1 and L_2, are given by:

$$L_1 : \boldsymbol{r}_1 = \begin{pmatrix} 5 \\ 4 \\ 2 \end{pmatrix} + s \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \text{ where } s \in \mathbb{R}$$

$$L_2 : \frac{x+1}{2} = y - 7 = \frac{z+5}{3}$$

(a) Show that the position vector of the point of intersection, L_1 and L_2 is $\begin{pmatrix} 1 \\ 8 \\ -2 \end{pmatrix}$. [3]

The plane Π contains the lines L_1 and L_2.

(b) (i) Write down the equation of Π, giving your answer in the form $\boldsymbol{r} = \boldsymbol{a} + \lambda\boldsymbol{b} + \mu\boldsymbol{c}$ where $\lambda, \mu \in \mathbb{R}$.

(ii) Given that $\boldsymbol{b} \times \boldsymbol{c} = \begin{pmatrix} -4 \\ -1 \\ 3 \end{pmatrix}$, show that the Cartesian equation of Π is $4x + y - 3z = 18$. [3]

The plane intersects the coordinate axes at $\mathrm{P}(4.5, 0, 0)$, $\mathrm{Q}(0, q, 0)$, and $\mathrm{R}(0, 0, r)$.

(c) Write down the value of

(i) q;

(ii) r. [2]

(d) Use a vector method to find the area of the triangle PQR. [5]

Another line, L_3, is normal to Π and passes through the point of intersection of L_1 and L_2.

(e) Write down an equation for L_3 in the form $\boldsymbol{r}_3 = \boldsymbol{m} + \gamma\boldsymbol{n}$. [2]

(f) Given that the point $\mathrm{S}(-11, 5, 7)$ lies on the line L_3, find γ. [2]

(g) Hence, find the volume of pyramid PQRS. [4]

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9. [Maximum mark: 8]

A line L_1 has vector equation $\boldsymbol{r} = \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix} + t \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ where $t \in \mathbb{R}$.

The plane Π_1 contains the line L_1 and passes through the point $(2, 1, 5)$.

(a) Show that the Cartesian equation of the plane Π_1 is $x + y - z = -2$. [4]

Consider the three planes

$$\Pi_1 : x + y - z = -2$$
$$\Pi_2 : 2x + by - z = 3$$
$$\Pi_3 : x - y + 2z = d$$

where $b, d \in \mathbb{Q}^+$.

The three planes intersect in a line.

(b) Find the value of b and the value of d. [4]

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Section B

Answer **all** questions in the answer booklet provided. Please start each question on a new page.

10. [Maximum mark: 17]

At Adam's Apple Orchard the weights of apples, W, in grams, are normally distributed with a mean 175 grams and standard deviation 8 grams.

(a) Find the probability that a randomly chosen apple weighs less than 170 grams. [2]

(b) It is found that 20% of the apples weigh more than w grams. Find w, correct to four significant figures. [2]

All orchards classify an apple as premium when its weight is between 170 and 185 grams.

(c) Find the percentage of apples that are classified as premium at Adam's Apple Orchard. [2]

After orders are completed, there are many apples left over. Boxes are filled with randomly chosen left-over apples. Each box contains 40 apples.

(d) Find the probability that a randomly chosen box contains at least 30 premium apples. [3]

(e) If 10 of these boxes are randomly selected, find the probability that exactly 4 boxes have at least 30 premium apples. [2]

At a neighbouring orchard the weights of apples, M, in grams, are normally distributed with mean μ and standard deviation σ. It is known that:

- 82% of their apples are classified as premium
- the percentage of apples that weigh less than 170 grams is twice the percentage of apples that weigh more than 185 grams.

(f) Find the value of μ. [6]

Do **not** write solutions on this page.

11. [Maximum mark: 15]

The function f is defined by $f(x) = e^{2x}(3x - 4)$, where $x \in \mathbb{R}$.

(a) Find $f'(x)$. [3]

(b) Hence or otherwise, find the coordinates of the point on the graph of $y = f(x)$ where the tangent is parallel to the line $y = x$. [3]

The region enclosed by the curve $y = f(x)$, the x-axis and the y-axis is rotated through 2π radians about the x-axis to form a solid of revolution.

(c) Find the volume of this solid. [4]

Consider a function g such that $g(0) = 1$ and $g'(0) = 2$.

(d) Find the value of

(i) $(f \circ g)(0)$;

(ii) $(f \circ g)'(0)$. [5]

12. [Maximum mark: 22]

Consider the points $\mathrm{A}(1, 2, 3)$, $\mathrm{B}(k, -2, 1)$ and $\mathrm{C}(5, 0, 2)$, where $k \in \mathbb{R}$.

(a) Write down $\vec{\mathrm{AB}}$ and $\vec{\mathrm{AC}}$. [2]

(b) Given that the points A, B and C lie on a straight line, show that $k = 9$. [1]

(c) For $k = 9$, let L_1 be the line passing through A, B and C.

(i) Find a vector equation of the line L_1.

(ii) Line L_2 has the equation $\frac{x-1}{2} = \frac{y}{3} = 1 - z$. Show that the lines L_1 and L_2 are skew. [10]

(d) For $k \neq 9$, let Π be the plane containing A, B and C.

(i) Find the Cartesian equation of the plane Π.

(ii) Find the coordinates of the point on the plane Π which is closest to the origin $(0, 0, 0)$. [9]

References:

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11. [Maximum mark: 21]

Three points $\mathrm{A}(3, 0, 0)$, $\mathrm{B}(0, -2, 0)$ and $\mathrm{C}(1, 1, -7)$ lie on the plane Π_1.

(a) (i) Find the vector $\vec{\mathrm{AB}}$ and the vector $\vec{\mathrm{AC}}$.

(ii) Hence find the equation of Π_1, expressing your answer in the form $ax + by + cz = d$, where $a, b, c, d \in \mathbb{Z}$. [7]

Plane Π_2 has equation $3x - y + 2z = 2$.

(b) The line L is the intersection of Π_1 and Π_2. Verify that the vector equation of L can be written as $\boldsymbol{r} = \begin{pmatrix} 0 \\ -2 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$. [2]

(c) The plane Π_3 is given by $2x - 2z = 3$. The line L and the plane Π_3 intersect at the point P.

(i) Show that at the point P, $\lambda = \frac{3}{4}$.

(ii) Hence find the coordinates of P. [3]

(d) The point $\mathrm{B}(0, -2, 0)$ lies on L.

(i) Find the reflection of the point B in the plane Π_3.

(ii) Hence find the vector equation of the line formed when L is reflected in the plane Π_3. [9]

References:

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12. [Maximum mark: 20]

Consider the non-zero vectors $\boldsymbol{u}$ and $\boldsymbol{v}$. Let θ be the angle between $\boldsymbol{u}$ and $\boldsymbol{v}$.

(a) Using the definitions of $\boldsymbol{u} \bullet \boldsymbol{v}$ and $\boldsymbol{u} \times \boldsymbol{v}$ in terms of $|\boldsymbol{u}|$, $|\boldsymbol{v}|$ and θ, show that $(\boldsymbol{u} \bullet \boldsymbol{v})^2 + |\boldsymbol{u} \times \boldsymbol{v}|^2 = |\boldsymbol{u}|^2 |\boldsymbol{v}|^2$. [2]

A triangle ABC has vertices $\mathrm{A}(0, 1, 2)$, $\mathrm{B}(p, q, 3)$ and $\mathrm{C}(3, 2, 1)$, $p, q \in \mathbb{Q}$.

The vectors $\boldsymbol{u}$ and $\boldsymbol{v}$ are defined as $\boldsymbol{u} = \overrightarrow{\mathrm{AB}}$ and $\boldsymbol{v} = \overrightarrow{\mathrm{AC}}$.

It is given that $\boldsymbol{u} \bullet \boldsymbol{v} = 3$ and the area of triangle ABC is $\sqrt{6}$.

(b) (i) Find the value of $|\boldsymbol{u} \times \boldsymbol{v}|$.

(ii) Hence, or otherwise, find the value of $|\boldsymbol{u}|$.

(iii) Hence, or otherwise, find the possible values of p and the corresponding values of q. [13]

Consider a new point D, the vector $\boldsymbol{w}$ is defined as $\boldsymbol{w} = \overrightarrow{\mathrm{CD}}$.

It is given that $\boldsymbol{u} \bullet \boldsymbol{w} = \boldsymbol{v} \bullet \boldsymbol{w} = 0$ and the area of triangle ACD is 5 square units.

(c) Assuming that $p = 1$, find the possible vectors for $\boldsymbol{w}$. [5]

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12. [Maximum mark: 18]

Line L is given by the vector equation $\boldsymbol{r}_1 = \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix} + s \begin{pmatrix} 2 \\ 3 \\ 6 \end{pmatrix}$ where $s \in \mathbb{R}$.

Line M is given by the vector equation $\boldsymbol{r}_2 = \begin{pmatrix} 9 \\ 9 \\ 11 \end{pmatrix} + t \begin{pmatrix} 4 \\ 1 \\ 2 \end{pmatrix}$ where $t \in \mathbb{R}$.

(a) Show that lines L and M intersect at a point A and find the position vector of A. [5]

(b) Verify that the lines L and M both lie in the plane Π given by $\boldsymbol{r} \bullet \begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix} = 7$. [3]

Point B has position vector $\begin{pmatrix} -3 \\ 12 \\ 2 \end{pmatrix}$. A line through B perpendicular to Π intersects Π at point C.

(c) (i) Find the position vector of C.

(ii) Hence, find $\left|\vec{\mathrm{BC}}\right|$. [7]

(d) Find the reflection of the point B in the plane Π. [3]

Do **not** write solutions on this page.

12. [Maximum mark: 18]

Line L is given by the vector equation $\boldsymbol{r}_1 = \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix} + s \begin{pmatrix} 2 \\ 3 \\ 6 \end{pmatrix}$ where $s \in \mathbb{R}$.

Line M is given by the vector equation $\boldsymbol{r}_2 = \begin{pmatrix} 9 \\ 9 \\ 11 \end{pmatrix} + t \begin{pmatrix} 4 \\ 1 \\ 2 \end{pmatrix}$ where $t \in \mathbb{R}$.

(a) Show that lines L and M intersect at a point A and find the position vector of A. [5]

(b) Verify that the lines L and M both lie in the plane Π given by $\boldsymbol{r} \bullet \begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix} = 7$. [3]

Point B has position vector $\begin{pmatrix} -3 \\ 12 \\ 2 \end{pmatrix}$. A line through B perpendicular to Π intersects Π at point C.

(c) (i) Find the position vector of C.

(ii) Hence, find $\left|\overrightarrow{\mathrm{BC}}\right|$. [7]

(d) Find the reflection of the point B in the plane Π. [3]

6. [Maximum mark: 8]

Consider the two planes

$$\Pi_1 : 2x - y + 2z = 6$$

$$\Pi_2 : 4x + 3y - z = 2$$

Let L be the line of intersection of Π_1 and Π_2.

(a) Verify that a vector equation of L is $\boldsymbol{r} = \begin{pmatrix} 0 \\ 2 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -2 \\ -2 \end{pmatrix}$, where $\lambda \in \mathbb{R}$. [3]

(b) Find the coordinates of the point P on L that is nearest to the origin. [5]

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7. [Maximum mark: 5]

A function f is defined as $f(x) = \arctan(x-2)$, where $2 \le x \le 2+\sqrt{3}$.

The region bounded by the curve, the y-axis, the x-axis and the line $y = \frac{\pi}{3}$ is rotated 360° about the y-axis to form a solid of revolution.

Find the volume of the solid.

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Do **not** write solutions on this page.

11. [Maximum mark: 20]

Two airplanes, A and B, have position vectors with respect to an origin O given respectively by

$$\boldsymbol{r}_A = \begin{pmatrix} 19 \\ -1 \\ 1 \end{pmatrix} + t \begin{pmatrix} -6 \\ 2 \\ 4 \end{pmatrix}$$

$$\boldsymbol{r}_B = \begin{pmatrix} 1 \\ 0 \\ 12 \end{pmatrix} + t \begin{pmatrix} 4 \\ 2 \\ -2 \end{pmatrix}$$

where t represents the time in minutes and $0 \leq t \leq 2.5$.

Entries in each column vector give the displacement east of O, the displacement north of O and the distance above sea level, all measured in kilometres.

(a) Find the three-figure bearing on which airplane B is travelling. [2]

(b) Show that airplane A travels at a greater speed than airplane B. [2]

(c) Find the acute angle between the two airplanes' lines of flight. Give your answer in degrees. [4]

The two airplanes' lines of flight cross at point P.

(d) (i) Find the coordinates of P.

(ii) Determine the length of time between the first airplane arriving at P and the second airplane arriving at P. [7]

Let $D(t)$ represent the distance between airplane A and airplane B for $0 \leq t \leq 2.5$.

(e) Find the minimum value of $D(t)$. [5]

Do **not** write solutions on this page.

12. [Maximum mark: 21]

The population, P, of a particular species of marsupial on a small remote island can be modelled by the logistic differential equation

$$\frac{\mathrm{d}P}{\mathrm{d}t} = kP\left(1 - \frac{P}{N}\right)$$

where t is the time measured in years and k, N are positive constants.

The constant N represents the maximum population of this species of marsupial that the island can sustain indefinitely.

(a) In the context of the population model, interpret the meaning of $\frac{\mathrm{d}P}{\mathrm{d}t}$. [1]

(b) Show that $\frac{\mathrm{d}^2P}{\mathrm{d}t^2} = k^2P\left(1 - \frac{P}{N}\right)\left(1 - \frac{2P}{N}\right)$. [4]

(c) Hence show that the population of marsupials will increase at its maximum rate when $P = \frac{N}{2}$. Justify your answer. [5]

(d) Hence determine the maximum value of $\frac{\mathrm{d}P}{\mathrm{d}t}$ in terms of k and N. [2]

Let P_0 be the initial population of marsupials.

(e) By solving the logistic differential equation, show that its solution can be expressed in the form

$$kt = \ln\frac{P}{P_0}\left(\frac{N - P_0}{N - P}\right).$$ [7]

After 10 years, the population of marsupials is $3P_0$. It is known that $N = 4P_0$.

(f) Find the value of k for this population model. [2]

References:

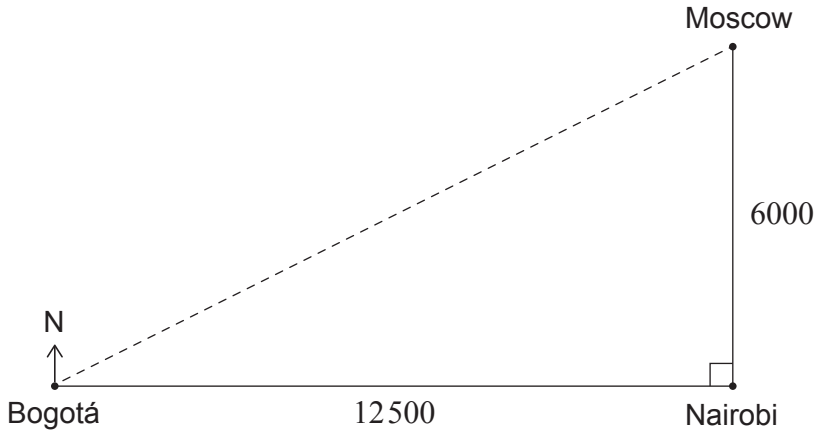
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2. [Maximum mark: 27]

The following question compares the distance and direction between cities on a flat surface to the distance and direction between cities on a sphere.

Consider a model where the cities of Bogotá, Moscow, and Nairobi lie on a flat surface. In this model, Nairobi is $6000\,\text{km}$ due south of Moscow and Bogotá is $12\,500\,\text{km}$ due west of Nairobi, as shown in the following diagram.

diagram not to scale



(a) (i) Find the distance from Bogotá to Moscow. [2]

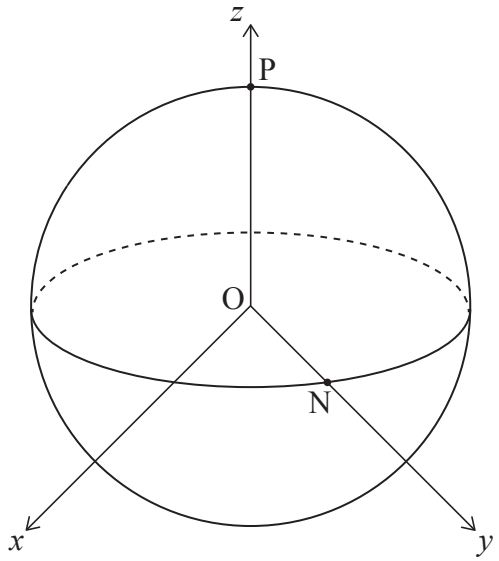
(ii) Find the bearing of Moscow from Bogotá. Give your answer in degrees. [3]

(This question continues on the following page)

(Question 2 continued)

In reality, these three cities lie on the curved surface of the Earth which will change the distances and directions found in part (a).

Now consider a curved model using a coordinate system (x, y, z) with its origin, O, at the centre of the Earth. The units of this system are thousands of kilometres and the Earth is modelled as a sphere with radius $6000\,\mathrm{km}$. The North Pole, P, lies on the z-axis, and Nairobi, N, is modelled as being on the equator and lying on the y-axis.



P has position vector $\overrightarrow{\mathrm{OP}} = \boldsymbol{p} = \begin{pmatrix} 0 \\ 0 \\ 6 \end{pmatrix}$ and N has position vector $\overrightarrow{\mathrm{ON}} = \boldsymbol{n} = \begin{pmatrix} 0 \\ 6 \\ 0 \end{pmatrix}$.

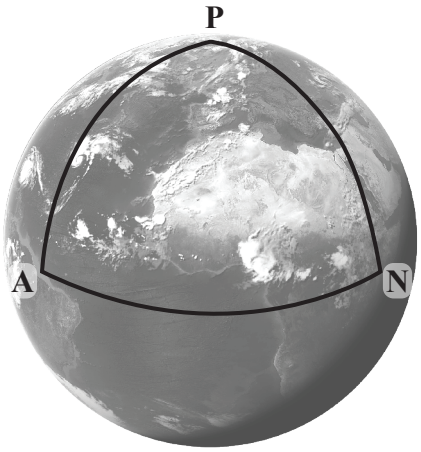
(b) (i) Use the scalar product to find the angle between $\boldsymbol{p}$ and $\boldsymbol{n}$. [2]

(ii) Show that the distance between P and N along the arc from P to N is $3000\pi\,\mathrm{km}$. [2]

(This question continues on the following page)

(Question 2 continued)

Point A, which is also on the equator, has position vector $\boldsymbol{a} = \begin{pmatrix} 6 \\ 0 \\ 0 \end{pmatrix}$ as shown in the following diagram.



P, N and A, and the arcs connecting them, form a spherical triangle.

The angle at vertex A is defined as the angle between the vectors $\boldsymbol{a} \times \boldsymbol{p}$ and $\boldsymbol{a} \times \boldsymbol{n}$.

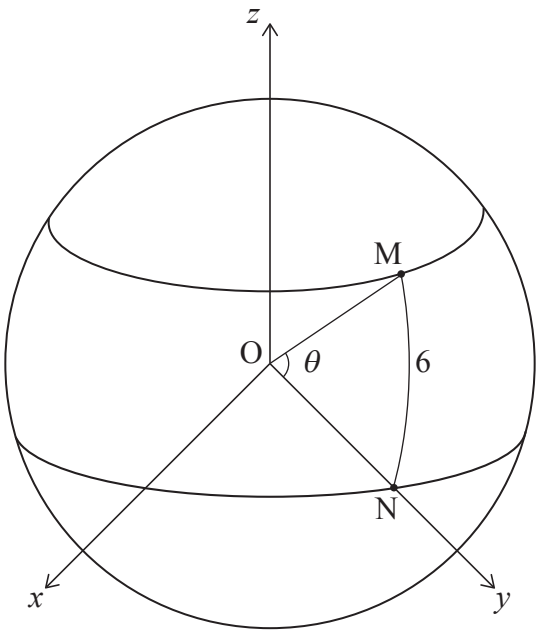
(c) (i) Find the vector $\boldsymbol{a} \times \boldsymbol{p}$. [2]

(ii) Show that the angle at vertex A in the spherical triangle is $90°$. [3]

(This question continues on the following page)

(Question 2 continued)

Moscow, M, has position vector $\overrightarrow{\mathrm{OM}} = \boldsymbol{m} = \begin{pmatrix} 0 \\ 6\cos\theta \\ 6\sin\theta \end{pmatrix}$, as shown in the following diagram.



The shortest distance between two points on the sphere lies along an arc of a circle on the sphere with centre O. In this model the shortest distance from Moscow to Nairobi is $6000\,\mathrm{km}$.

(d) Show that $\theta = 57.3°$, correct to three significant figures. [2]

Bogotá, B, is west of Nairobi and has position vector $\overrightarrow{\mathrm{OB}} = \boldsymbol{b} = \begin{pmatrix} 6\sin 120^\circ \\ 6\cos 120^\circ \\ 0 \end{pmatrix}$.

(e) Find the shortest distance from Bogotá to Moscow on the sphere. [5]

The bearing from B to M is defined as the angle at vertex B in the spherical triangle containing B, M and P. It is given that $\boldsymbol{b} \times \boldsymbol{p} = \begin{pmatrix} 36\cos 120^\circ \\ -36\sin 120^\circ \\ 0 \end{pmatrix}$.

(f) Using the method from part (c), find the bearing from Bogotá to Moscow. [6]

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References:

2(c) Google Maps/Google Earth. Data from Data SIO, NOAA, U.S. Navy, NGA, GEBCOLandsat / Copernicus, IBCAO, U.S. Geological Survey, PGC/NASA. Imagery from 14/12/2015. Image available at: https://earth.google.com/web/@12.01529518,-18.56070747,-158.39383184a,23597813.93249989d,30.00008083y,359.99981502h,0t,0r/data=CgRCAggBOgMKATBCAggASg0I_____ARAA.

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